CHEMICAL ENGINEERING CLASS PROJECT REPORT

Finite Difference Solution & Thermal Adaptation Audit: Exercise 5-34
Course: Numerical Methods in Heat Transfer | Absolute Temperature Version (273.15 K Offset)

MANUAL AUDIT GUIDELINE & LEGEND: Per project rubric (092126.pdf), all items tagged with an "@ [AUDIT]" symbol represent critical parameters, algebraic finite difference equations, matrix entries, numerical node temperatures, and compliance decisions that must be audited and verified BY HAND by the student pair prior to final report certification and GitHub repository submission. All thermal values below are evaluated in absolute temperature (Kelvin, with 273.15 K offset).

1. Problem Statement & System Constraints

System Configuration: A nonmetal plate with thickness L_2 = 20 mm is attached to the upper surface of an ASME SB-96 copper-silicon plate with thickness L_1 = 30 mm. The composite wall operates under 1D steady-state heat conduction in absolute temperature (Kelvin).

@ [AUDIT] • Plate 1 Properties: L_1 = 30 mm = 0.030 m, k_1 = 36.0 W/m·K (ASME SB-96 Copper-Silicon Plate)

@ [AUDIT] • Plate 2 Properties (Base): L_2 = 20 mm = 0.020 m, k_2 = 0.050 W/m·K (Nonmetal Outer Insulation)

@ [AUDIT] • Bottom Surface (x = 0 mm): q_o = 150.0 W/m² uniform specified flux applied at x = 0 mm.

@ [AUDIT] • Top Surface (x = 50 mm): Exposed to ambient air at T_∞ = 15.0 °C = 288.15 K with base convection coefficient h = 5.0 W/m²·K.

@ [AUDIT] • Code Safety Criterion: Per ASME BPVC.IV-2015 (HF-300), the maximum temperature in the ASME SB-96 plate must not exceed T_max,limit = 93.0 °C = 366.15 K.

Cooling Modifications Objective: To comply with the project rubric (092126.pdf), two thermal adaptation modifications designed for faster cooling are evaluated alongside the base problem:

@ [AUDIT] • Modification 1 (Enhanced Convection): Forced convective air cooling on the top surface with h = 25.0 W/m²·K (5× increase).

@ [AUDIT] • Modification 2 (Conductive Cladding): Replacing nonmetal insulation with high-conductivity metal cladding with k_2 = 80.2 W/m·K.

1.1 Fundamental Physical Assumptions & Modeling Approximations

@ [AUDIT] 1D Steady-State Heat Conduction: Heat transfer occurs exclusively in one dimension along the wall thickness (x-direction); transverse thermal gradients and edge heat losses are assumed negligible ($\partial T/\partial y = \partial T/\partial z = 0$, $\partial T/\partial t = 0$).

@ [AUDIT] Constant Thermal Properties: Material conductivities for both the ASME SB-96 copper-silicon plate (k_1 = 36.0 W/m·K) and outer nonmetal/cladding layer (k_2 = 0.050 W/m·K or 80.2 W/m·K) remain uniform and temperature-independent.

@ [AUDIT] Zero Internal Heat Generation: No volumetric internal thermal energy generation or chemical reaction heating exists within either material domain ($\dot{e}_{gen} = 0$).

@ [AUDIT] Perfect Thermal Contact at Interface: Ideal contact is assumed at the material interface (x = 30 mm, Node 6) with zero thermal contact resistance ($R_{t,c} = 0$), ensuring temperature and heat flux continuity.

@ [AUDIT] Uniform Boundary Flux & Convection: The bottom surface (x = 0 mm) experiences a constant, uniform heat flux (q_0 = 150.0 W/m²), and the top surface (x = 50 mm) is exposed to a uniform ambient fluid (T_∞ = 288.15 K / 15.0 °C) with a constant convection coefficient (h = 5.0 or 25.0 W/m²·K).

2. Nodal Discretization & Grid Geometry

@ [AUDIT] Discretization Scheme: The total thickness L = L_1 + L_2 = 50 mm is discretized using a uniform nodal spacing of Δx = 5.0 mm = 0.005 m. This divides the composite medium into N = 11 nodes (Node 0 through Node 10).

Node #	Position x (mm)	Material Assignment	Control Volume Description
Node 0	0.0 mm	ASME SB-96 Plate	Bottom surface, specified flux q_0 @ [AUDIT]
Node 1	5.0 mm	ASME SB-96 Plate	Interior metal node @ [AUDIT]
Node 2	10.0 mm	ASME SB-96 Plate	Interior metal node @ [AUDIT]
Node 3	15.0 mm	ASME SB-96 Plate	Interior metal node @ [AUDIT]
Node 4	20.0 mm	ASME SB-96 Plate	Interior metal node @ [AUDIT]
Node 5	25.0 mm	ASME SB-96 Plate	Interior metal node @ [AUDIT]
Node 6	30.0 mm	Material Interface	Contact boundary (k_1, k_2) @ [AUDIT]
Node 7	35.0 mm	Outer Layer Plate	Interior outer node @ [AUDIT]
Node 8	40.0 mm	Outer Layer Plate	Interior outer node @ [AUDIT]
Node 9	45.0 mm	Outer Layer Plate	Interior outer node @ [AUDIT]
Node 10	50.0 mm	Outer Layer Surface	Top surface, convection (h, T_∞) @ [AUDIT]

3. Finite Difference Formulations & Matrix Systems

Derivation Principles: The algebraic finite difference equations are derived from steady 1D energy balances (∑ E_in = 0) applied to each control volume. Students must manually audit each nodal equation before linear solver execution.

3.1 Nodal Governing Equations (General Form)

@ [AUDIT] • Bottom Boundary (Node 0, Flux Boundary): $-T_0 + T_1 = -(q_0 \cdot \Delta x / k_1) = -(150 \times 0.005 / 36) = -0.020833$ K

@ [AUDIT] • Interior Metal Nodes (Nodes 1–5): T_{m-1} - 2T_m + T_{m+1} = 0 (for m = 1, 2, 3, 4, 5)

@ [AUDIT] • Interface Boundary (Node 6): $k_1 T_5 - (k_1 + k_2) T_6 + k_2 T_7 = 0 \Rightarrow 36 T_5 - (36 + k_2) T_6 + k_2 T_7 = 0$

@ [AUDIT] • Interior Outer Nodes (Nodes 7–9): T_{m-1} - 2T_m + T_{m+1} = 0 (for m = 7, 8, 9)

@ [AUDIT] • Top Convection Boundary (Node 10): $T_9 - (1 + h \cdot \Delta x / k_2) T_{10} = -(h \cdot \Delta x / k_2) T_\infty$

3.2 Linear Algebraic Systems for All Three Cases

a) Base Problem System Matrix: For h = 5.0 W/m²·K, k_1 = 36.0 W/m·K, k_2 = 0.050 W/m·K, T_∞ = 288.15 K:

@ [AUDIT] • Node 0 Equation: $-T_0 + T_1 = -0.020833$ K

@ [AUDIT] • Nodes 1–5 Eqns: T_{m-1} - 2T_m + T_{m+1} = 0

@ [AUDIT] • Node 6 Equation: $36 T_5 - 36.05 T_6 + 0.05 T_7 = 0$

@ [AUDIT] • Nodes 7–9 Eqns: T_{m-1} - 2T_m + T_{m+1} = 0

@ [AUDIT] • Node 10 Equation: $T_9 - 1.5\ T_{10} = -144.075000$ K (since $-(h \cdot \Delta x/k_2)\ T_\infty = -0.5 \times 288.15$ K = -144.075 K)

b) Modification 1 (Enhanced Convection, h = 25 W/m²·K) System Matrix: For h = 25.0 W/m²·K, k_1 = 36.0 W/m·K, k_2 = 0.050 W/m·K, T_∞ = 288.15 K:

@ [AUDIT] • Node 0–9 Eqns: Identical to Base Case equations.

@ [AUDIT] • Node 10 Equation: $T_9 - 3.5\ T_{10} = -720.375000$ K (since $-(h \cdot \Delta x/k_2)\ T_\infty = -2.5 \times 288.15$ K = -720.375 K)

c) Modification 2 (Conductive Cladding, k_2 = 80.2 W/m·K) System Matrix: For h = 5.0 W/m²·K, k_1 = 36.0 W/m·K, k_2 = 80.2 W/m·K, T_∞ = 288.15 K:

@ [AUDIT] • Node 0–5 Eqns: Identical to Base Case equations.

@ [AUDIT] • Node 6 Equation: $36\ T_5 - 116.2\ T_6 + 80.2\ T_7 = 0$ (since 36 + 80.2 = 116.2)

@ [AUDIT] • Nodes 7–9 Eqns: T_{m-1} - 2T_m + T_{m+1} = 0

@ [AUDIT] • Node 10 Equation: $T_9 - 1.0003117\ T_{10} = -0.0898223$ K (since $-(h \cdot \Delta x/k_2)\ T_\infty = -(5 \times 0.005/80.2) \times 288.15$ K = -0.0898223 K)

4. Comprehensive Numerical Results & Comparison

Nodal Temperature Summary Table: Below is the complete node-by-node temperature distribution across the three evaluated system configurations in absolute Kelvin (with Celsius equivalents). Students must hand-audit these values against analytical solutions.

Node	x (mm)	Base Case (K / °C) (h=5, k_2=0.05)	Mod 1: Convection (K / °C) (h=25, k_2=0.05)	Mod 2: Cladding (K / °C) (h=5, k_2=80.2)
0 (Bottom)	0.0	378.2750 K (105.1250 °C) @	354.2750 K (81.1250 °C) @	318.3124 K (45.1624 °C) @
1	5.0	378.2542 K (105.1042 °C) @	354.2542 K (81.1042 °C) @	318.2916 K (45.1416 °C) @
2	10.0	378.2333 K (105.0833 °C) @	354.2333 K (81.0833 °C) @	318.2707 K (45.1207 °C) @
3	15.0	378.2125 K (105.0625 °C) @	354.2125 K (81.0625 °C) @	318.2499 K (45.0999 °C) @
4	20.0	378.1917 K (105.0417 °C) @	354.1917 K (81.0417 °C) @	318.2291 K (45.0791 °C) @
5	25.0	378.1708 K (105.0208 °C) @	354.1708 K (81.0208 °C) @	318.2082 K (45.0582 °C) @
6 (Interface)	30.0	378.1500 K (105.0000 °C) @	354.1500 K (81.0000 °C) @	318.1874 K (45.0374 °C) @
7	35.0	363.1500 K (90.0000 °C) @	339.1500 K (66.0000 °C) @	318.1781 K (45.0281 °C) @
8	40.0	348.1500 K (75.0000 °C) @	324.1500 K (51.0000 °C) @	318.1687 K (45.0187 °C) @
9	45.0	333.1500 K (60.0000 °C) @	309.1500 K (36.0000 °C) @	318.1594 K (45.0094 °C) @
10 (Top)	50.0	318.1500 K (45.0000 °C) @	294.1500 K (21.0000 °C) @	318.1500 K (45.0000 °C) @

4.1 ASME BPVC Code Compliance Summary

Case Configuration	Max SB-96 Temp (T_0)	ASME Limit (T_max)	BPVC Compliance Status	Safety Margin
Base Case (h=5, k_2=0.05)	378.28 K (105.13 °C) @ [AUDIT]	≤ 366.15 K @	NON-COMPLIANT @	-12.13 K (Violation) @

Mod 1: Convection (h=25, k_2=0.05)	354.28 K (81.13 °C) @ [AUDIT]	≤ 366.15 K @	**COMPLIANT @**	+11.88 K Below Limit @
Mod 2: Cladding (h=5, k_2=80.2)	318.31 K (45.16 °C) @ [AUDIT]	≤ 366.15 K @	**COMPLIANT @**	+47.84 K Below Limit @

5. Manual Analytical Verification & Checks

Mathematical Verification Principle: Per project rubric requirement, numerical finite difference solutions must be verified against exact 1D steady-state heat conduction formulas derived from thermal resistance network theory in absolute temperature.

@ [AUDIT] • General 1D Thermal Resistance Network Formula: $T_0 = T_{\infty} + q_0 \cdot [(1 / h) + (L_2 / k_2) + (L_1 / k_1)]$

5.1 Analytical Hand-Audit Computations

@ [AUDIT] a) Base Case Hand Check: $T_0 = 288.15 \text{ K} + 150 \cdot [(1 / 5) + (0.020 / 0.05) + (0.030 / 36)] = 288.15 + 150 \cdot [0.20 + 0.40 + 0.0008333] = 288.15 + 150 \cdot (0.6008333) = 378.2750 \text{ K}$ (105.1250 °C)

@ [AUDIT] b) Modification 1 Hand Check: $T_0 = 288.15 \text{ K} + 150 \cdot [(1 / 25) + (0.020 / 0.05) + (0.030 / 36)] = 288.15 + 150 \cdot [0.04 + 0.40 + 0.0008333] = 288.15 + 150 \cdot (0.4408333) = 354.2750 \text{ K}$ (81.1250 °C)

@ [AUDIT] c) Modification 2 Hand Check: $T_0 = 288.15 \text{ K} + 150 \cdot [(1 / 5) + (0.020 / 80.2) + (0.030 / 36)] = 288.15 + 150 \cdot [0.20 + 0.00024938 + 0.0008333] = 288.15 + 150 \cdot (0.2010826) = 318.3124 \text{ K}$ (45.1624 °C)

5.2 Calculation of Minimum Convection Coefficient (h_min)

@ [AUDIT] Hand Derivation of h_min: To enforce ASME BPVC compliance ($T_0 \le 366.15$ K / 93.0 °C) under base insulation ($k_2 = 0.05$ W/m·K):

@ [AUDIT] • Step 1 Equation: $T_0 = 288.15 + 150 / h + 150 \cdot (0.40 + 0.0008333) \le 366.15 \text{ K} \Rightarrow 288.15 + 150 / h + 60.125 \le 366.15 \text{ K}$

@ [AUDIT] • Step 2 Isolation: $348.2750 + 150 / h \le 366.15 \text{ K} \Rightarrow 150 / h \le 17.8750 \text{ K}$

@ [AUDIT] • Step 3 Final h_min Result: $h \ge 150 / 17.8750 = 8.391608 \text{ W/m}^2\text{·K} \Rightarrow h_{min} = 8.3916 \text{ W/m}^2\text{·K}$ (≈ 8.39 W/m²·K)

6. Student Pair Audit Checklist & Certification Block

Peer Verification Protocol: Peer Verification Protocol: Before final submission, the student pair must physically place a checkmark in each item below and sign the certification block as mandated by the course guidelines.

Check	Audit Component	Required Manual Verification Task
[ ]	1. Boundary & Material Inputs @	Verify k_1=36, k_2=0.05/80.2, L_1=30mm, L_2=20mm, q_0=150, T_{∞}=288.15K (15°C), h=5/25.
[ ]	2. Nodal Discretization Grid @	Verify Δx=5mm spacing and 11-node positioning from x=0 to x=50mm.
[ ]	3. Finite Difference Equations @	Verify control volume energy balances for Node 0, Node 6, and Node 10.
[ ]	4. Matrix Formulation Coefficients @	Audit row entries for Base Case, Mod 1 (h=25), and Mod 2 (k_2=80.2).
[ ]	5. Numerical Solvers & Profile @	Cross-check FDM output node temperatures against manual calculations in Kelvin.

[ ]	6. ASME BPVC Compliance Decision @	Confirm T_max ≤ 366.15 K (93 °C) threshold checks for Base (378.28 K), Mod 1, and Mod 2.
[ ]	7. Minimum Convection h_min @	Manually calculate and verify h_min = 8.3916 W/m²·K threshold.

JOINT HAND-SIGNED CERTIFICATION OF MANUAL AUDIT:

"We hereby certify that we have worked as entry-level chemical engineers to solve, adapt, and audit Exercise 5-34. All calculations marked with '@ [AUDIT]' in this report have been verified by hand against fundamental heat transfer laws in absolute temperature (273.15 K offset), and all prompt engineering steps and AI iterations have been logged in our GitHub repository per course requirements."

Student 1 Signature: LJPP Date: 10/26/26 Name: Lorena J. Pérez

Student 2 Signature: Anelisse Torres Date: 10/26/26 Name: Anelisse M. Torres